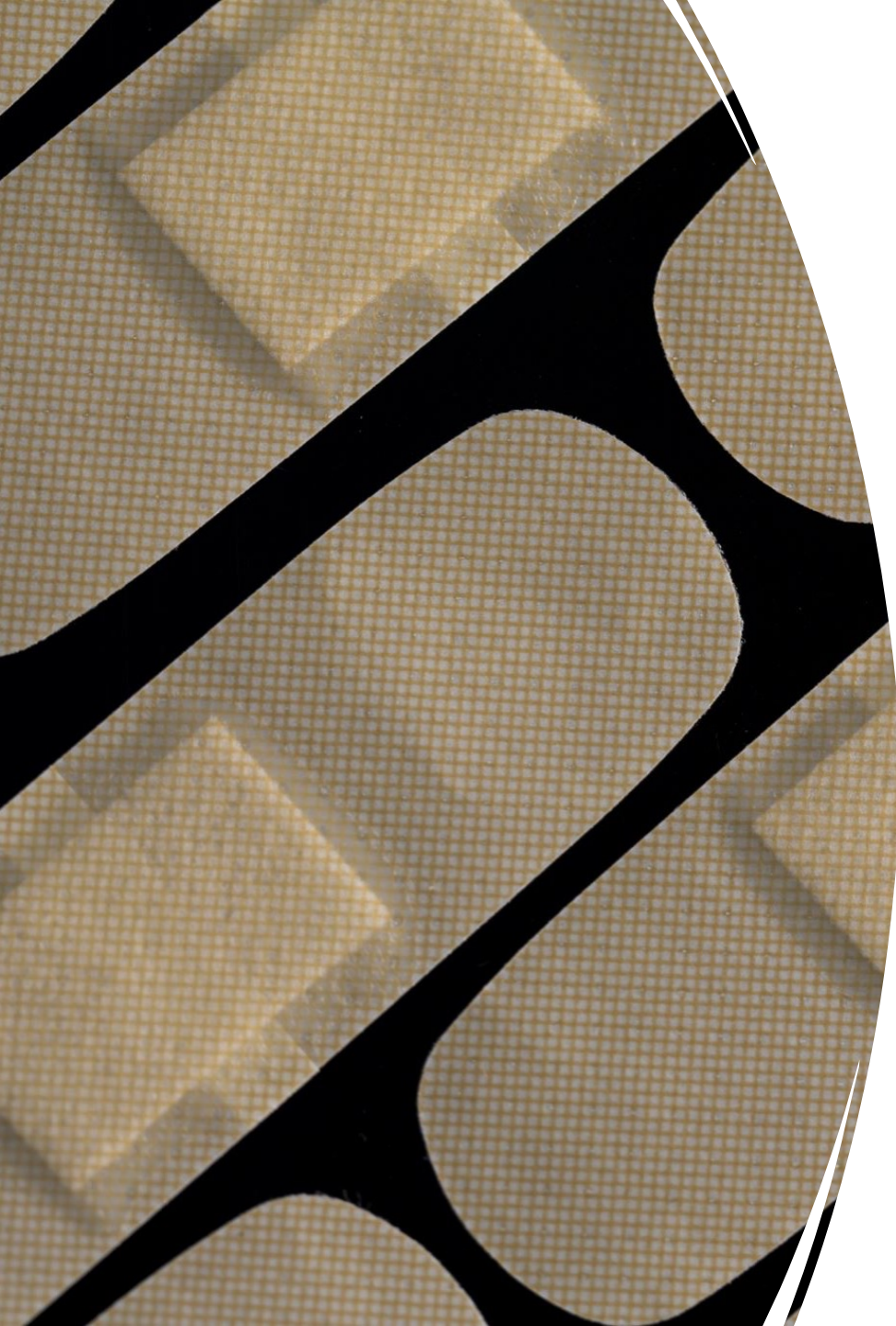




WCS 82

Painful Red Joint Monoarthropathy

Dr Carmen Ho

Outline of lecture

- Approach to patients with monoarthritis
 - Septic arthritis
 - Crystal arthritis
 - Haemarthrosis

Diseases causing monoarticular symptoms

Septic

- **Bacterial**
- **Mycobacterial**
- **Lyme disease**

Crystal deposition diseases

- Gout
- Calcium pyrophosphate deposition (CPPD) disease (pseudogout)
- Hydroxyapatite deposition disease
- Calcium oxalate deposition disease

Trauma

- Fracture
- ACL/PCL tear
- Haemarthrosis

Others

- Osteoarthritis
- Juvenile idiopathic arthritis (JIA)
- Coagulopathy
- Avascular necrosis of bone

Polyarticular diseases occasionally presenting with a monoarticular onset

- Rheumatoid arthritis (RA)
- Juvenile idiopathic arthritis (JIA)
- Spondyloarthritis (SpA)
- Psoriatic arthritis (PsA)
- Reactive arthritis
- Viral arthritis



Approach to patients with monoarticular problem

- Careful history taking
- Careful physical examination
- Relevant investigations

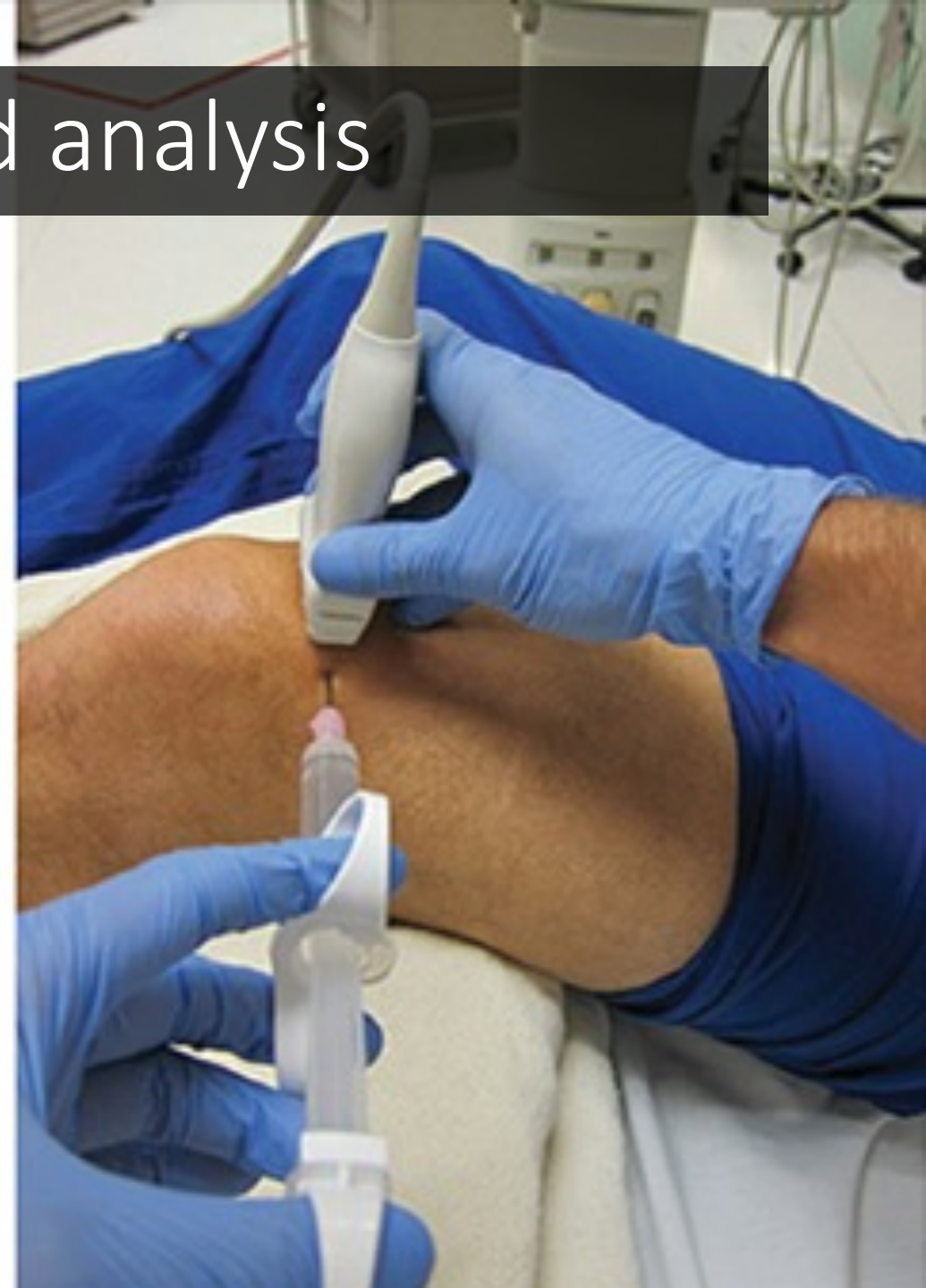
History – important clues

Clinical information	Possible diagnoses
• Pain coming on in seconds or minutes	Fracture, internal derangement
• Pain coming on over several hours or 1 to 2 days	Infection, crystal arthritis
• Pain coming on over days to weeks	Inflammatory arthritis syndromes, indolent infection, osteoarthritis
• Joint overused or damaged, either recently or in the past	Traumatic causes, osteoarthritis
• Previous acute attacks of self limiting arthritis	Crystal arthritis, inflammatory arthritis syndromes
• Treatment with immunosuppressive agents e.g. steroid, cytotoxic agent	Infection, osteonecrosis
• History of bleeding disorders or treatment with anti-coagulants	Haemoarthrosis

Relevant investigations



Synovial (joint) fluid analysis



Synovial fluid analysis

INDICATIONS

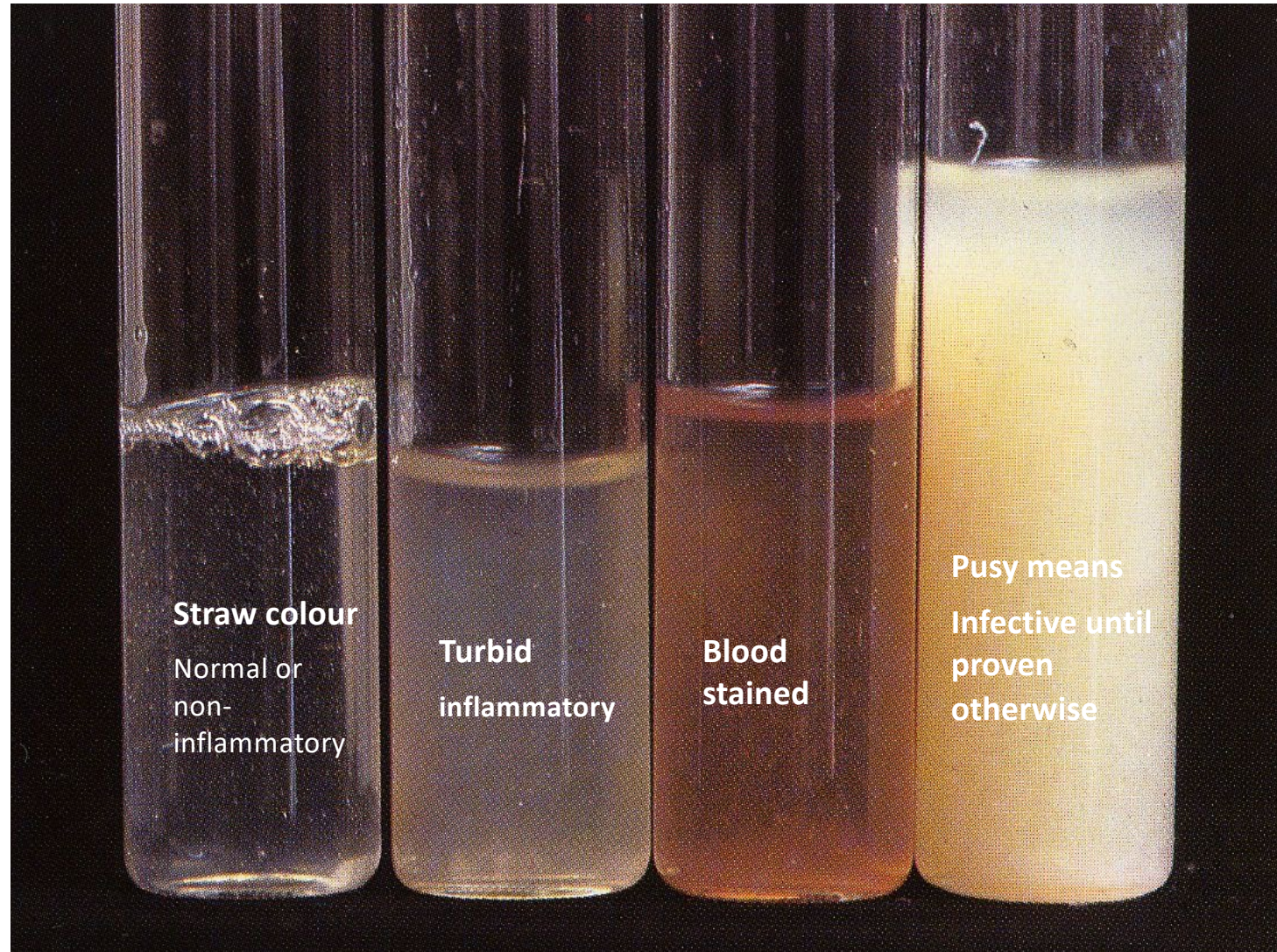
- Suspicion of infection
- Suspicion of crystal induced arthritis
- Suspicion of haemarthrosis
- Differentiating inflammatory from non-inflammatory arthritis

Examination of synovial fluid

Naked eye appearance	Colour Turbidity Viscosity
Microscopy	• Wet films • Cell stains and counts • Polarised light
Microbiology	• Gram stain • Culture • Special techniques e.g. ZN stain, fungal stains

Classification of synovial effusions

	Non-inflammatory (Normal /OA)	Inflammatory (RA/Crystal)	Septic
Viscosity	High	Low	Variable
Colour	Straw to yellow	Yellow	Variable
Clarity	Transparent	Translucent	Opaque
WBC (per mm ³)	200 to 2000	2000 to 75000	Often >100,000
PMN leukocytes (%)	<25	Often >50	>75



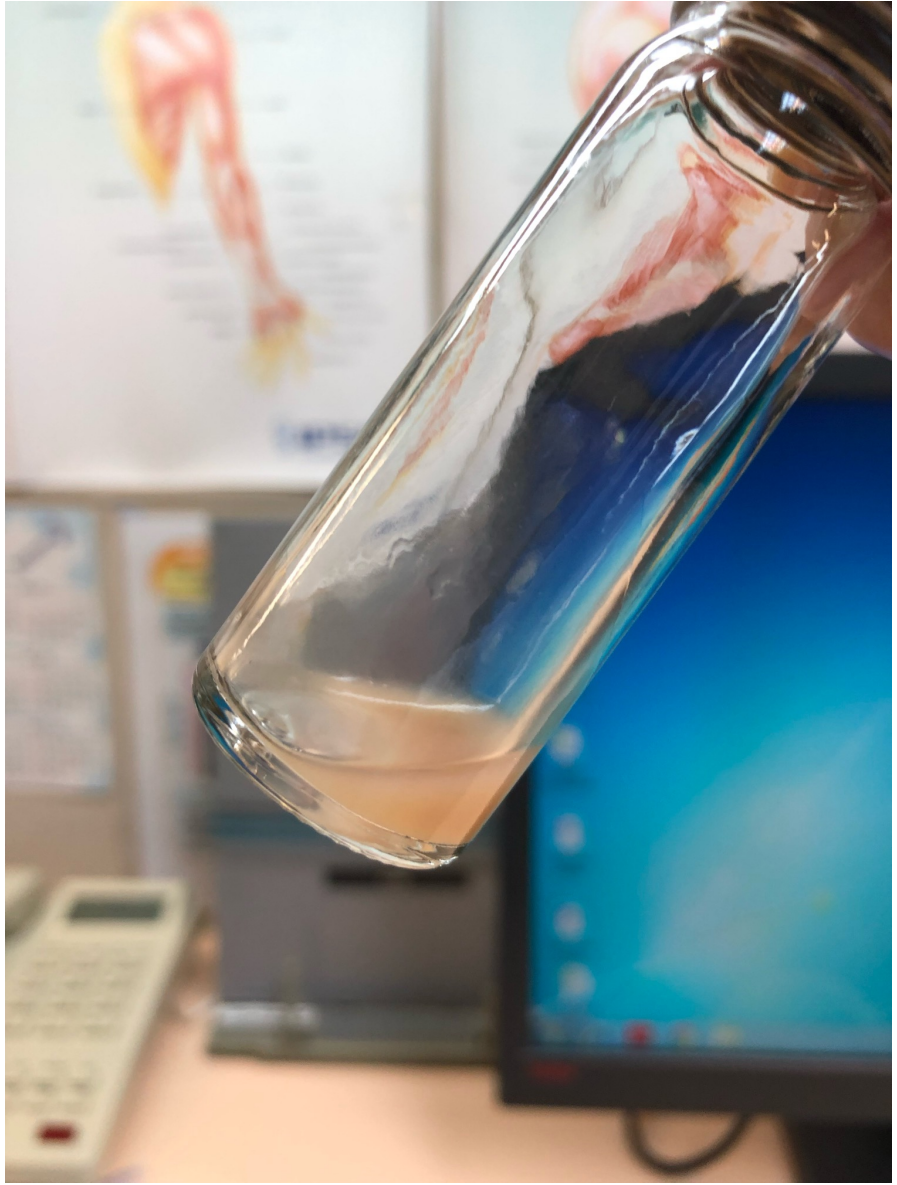
Straw colour

Normal or
non-
inflammatory

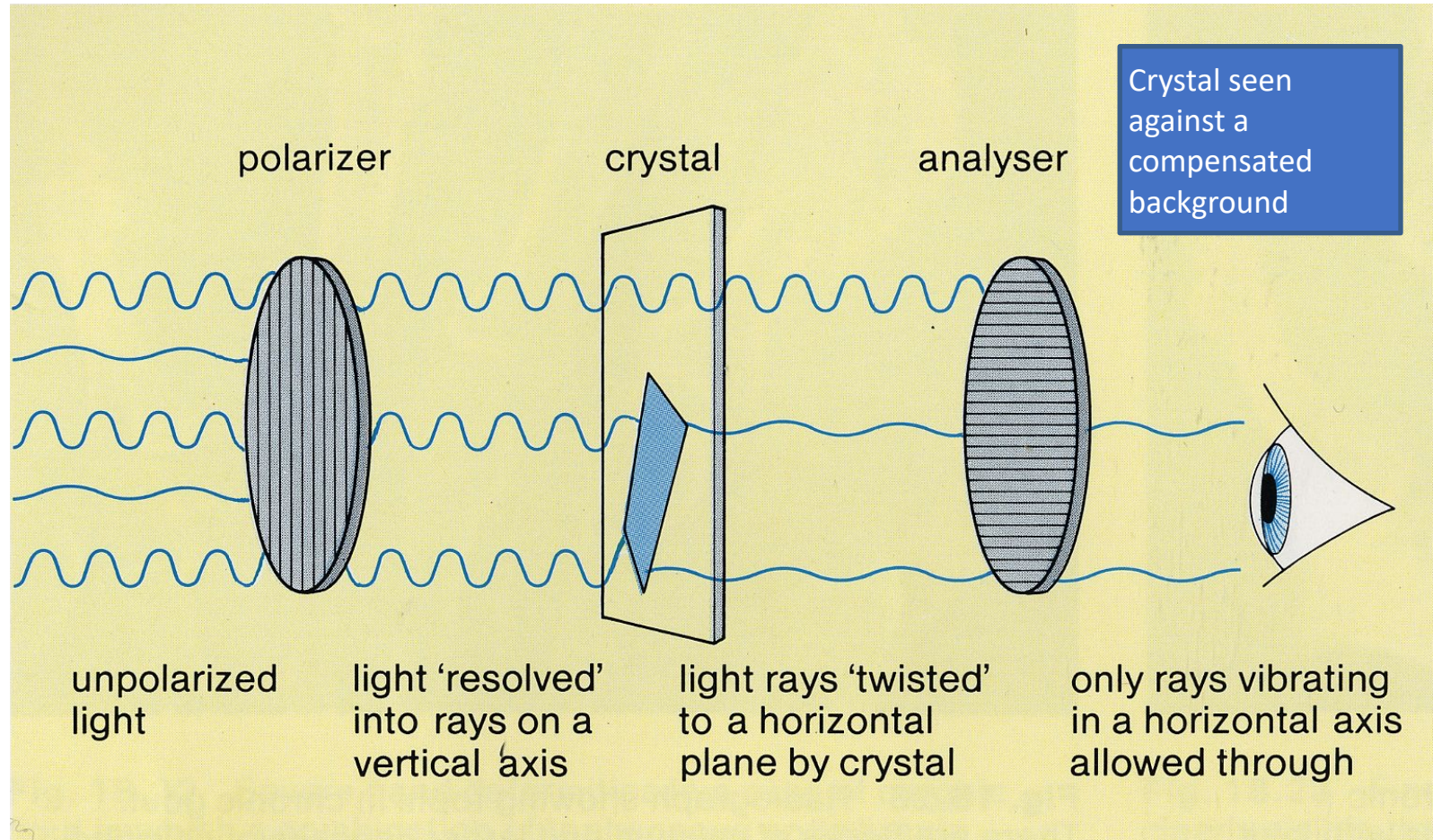
Turbid
inflammatory

Blood
stained

Pusy means
Infective until
proven
otherwise

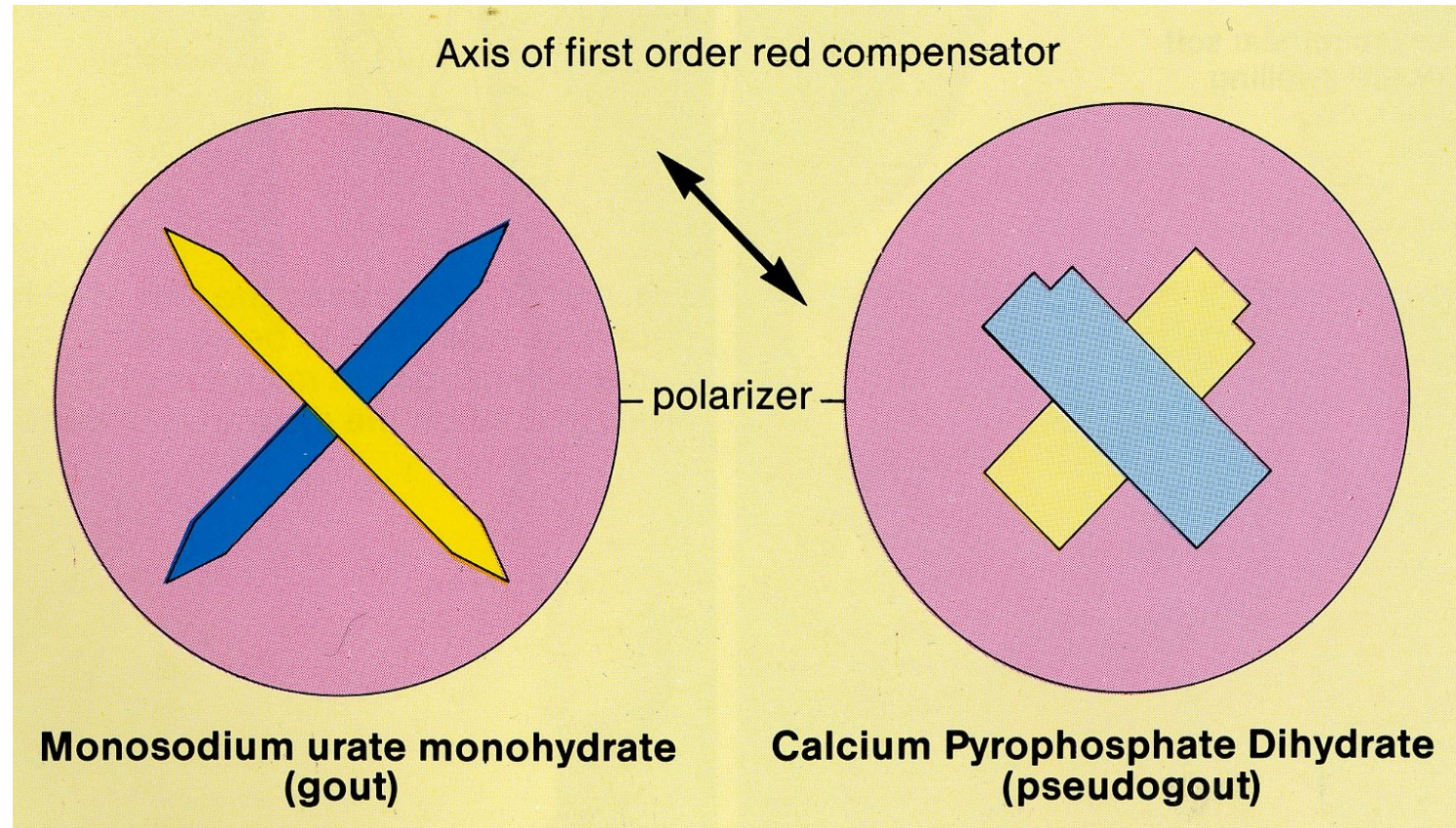


Polarised microscopy

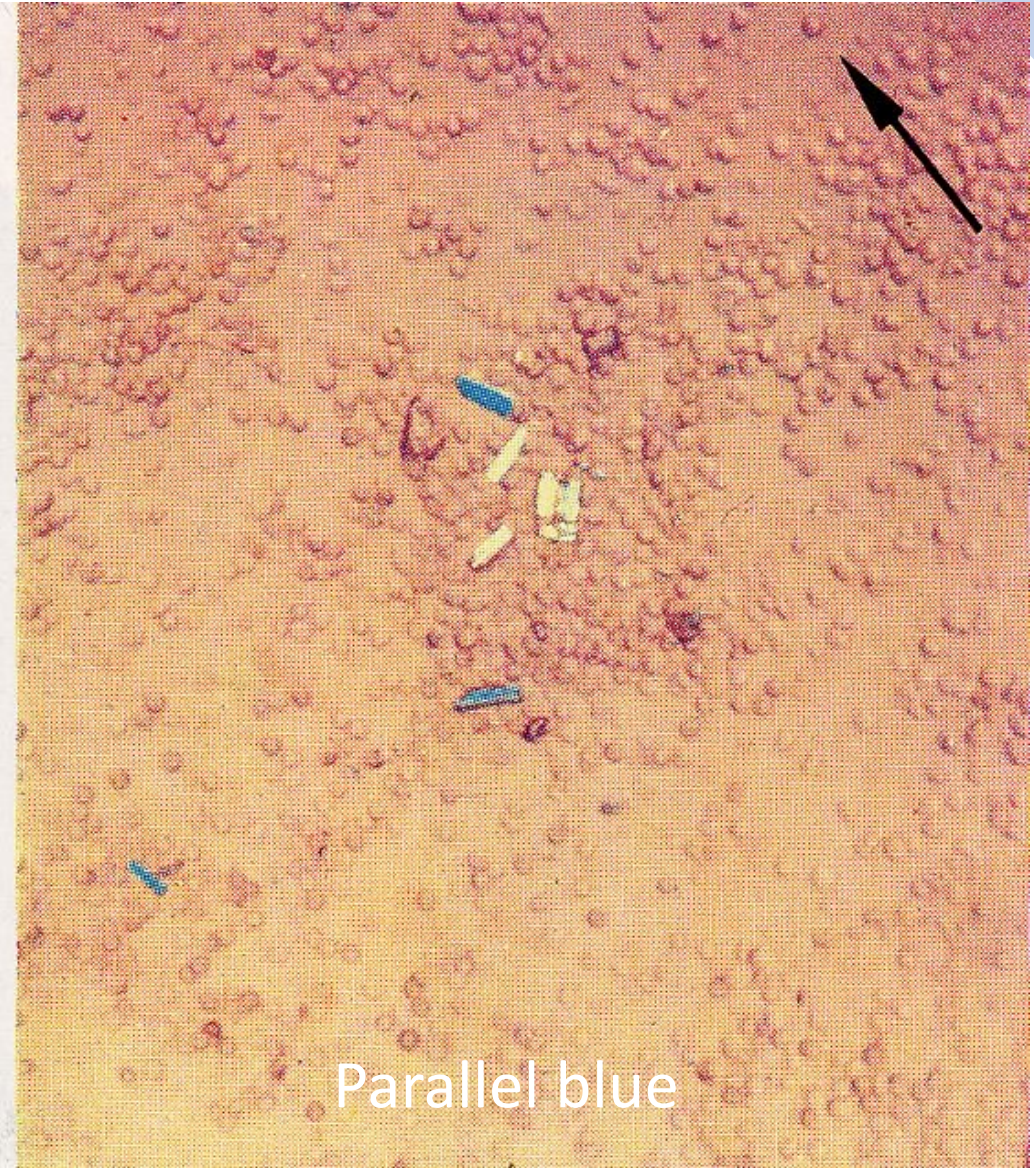
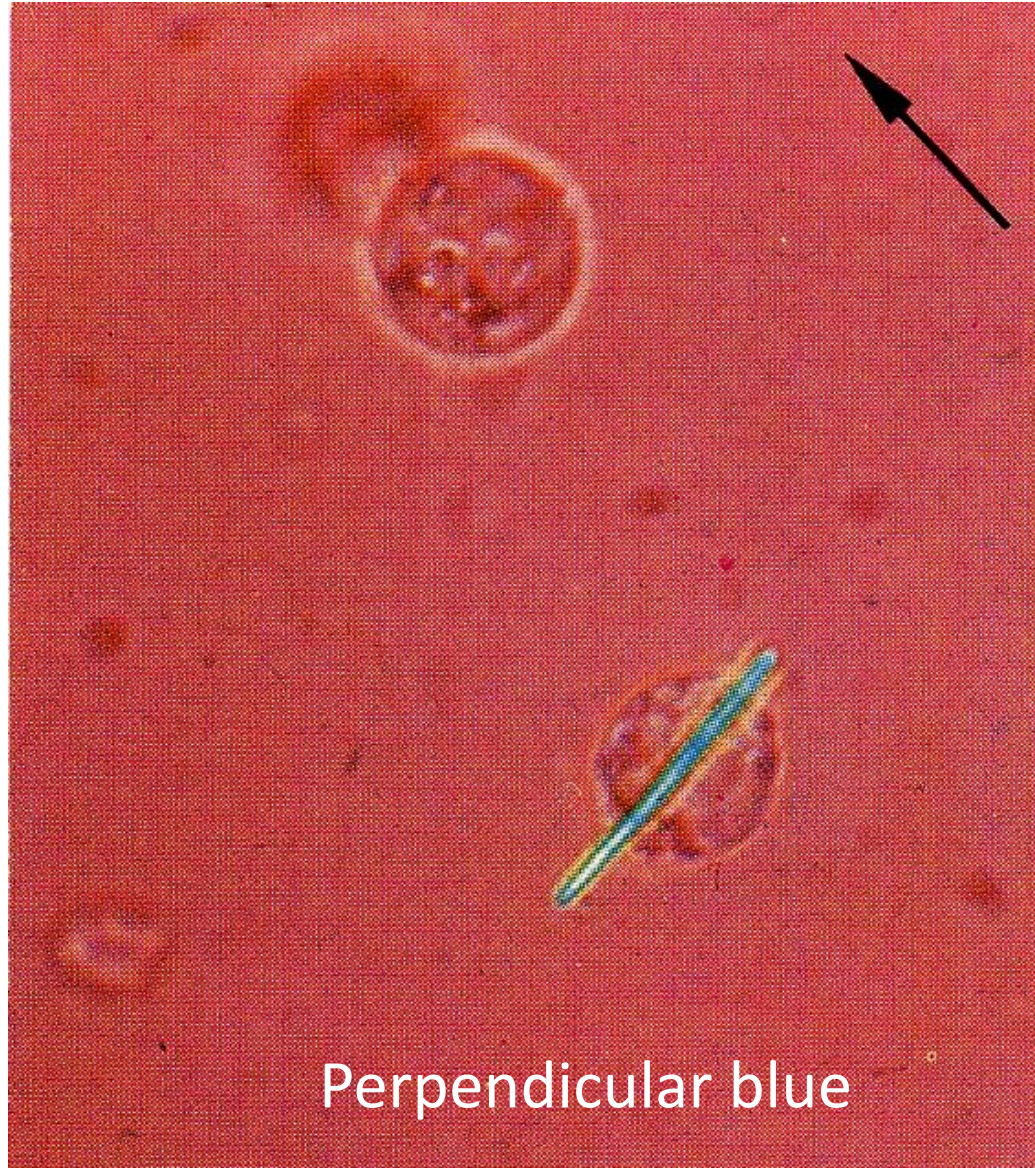


A birefringent crystal altering the vector of plane of polarised light, allowing it to be seen at the microscope eye piece

Negative and positive birefringence



Birefringence can be detected by a colour shift to blue (positive) or yellow (negative) when the long axis of the crystal is aligned with the optical axis of a first order red compensator



Other diagnostic tools for monoarthritis

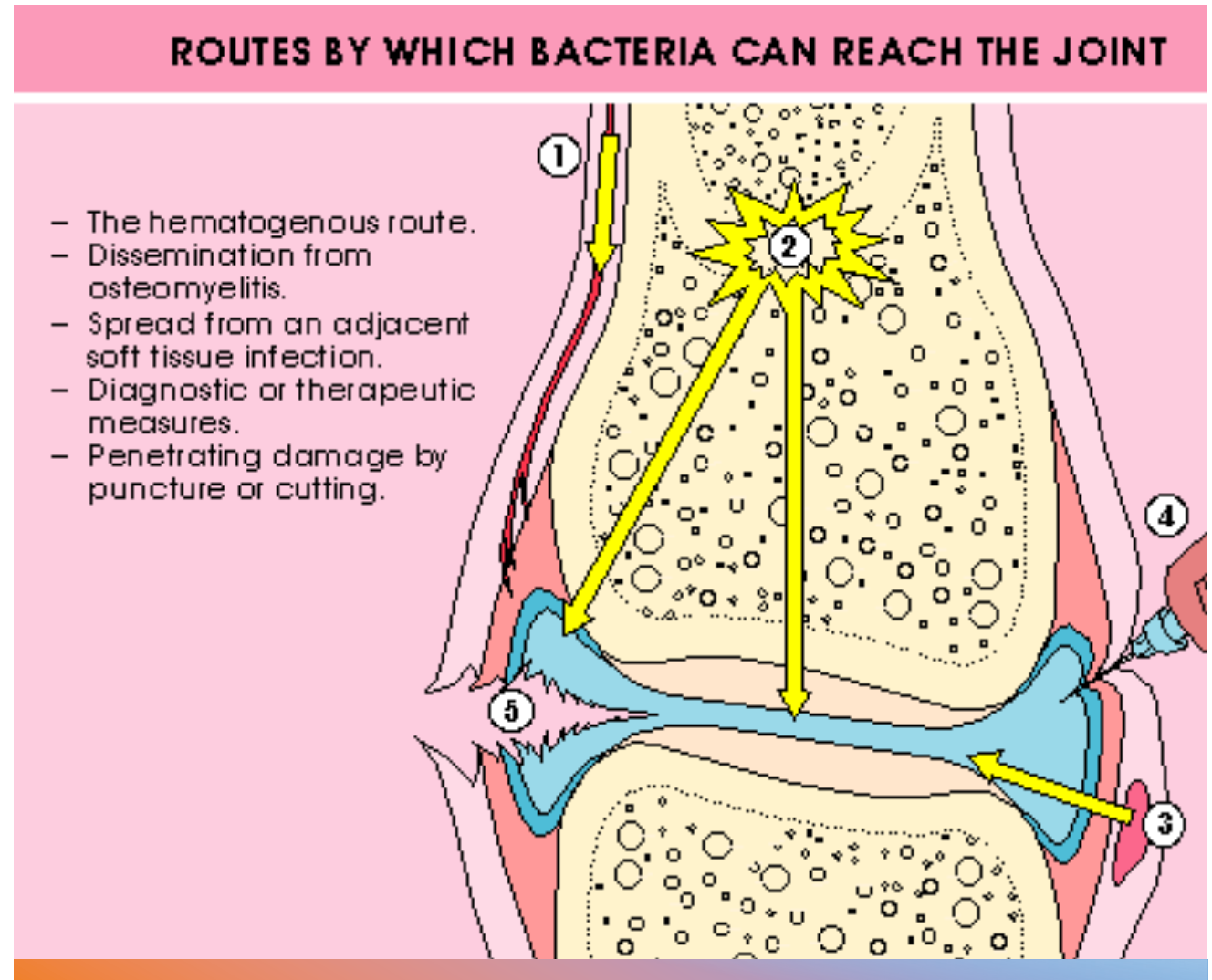
- Complete blood count
- Erythrocyte sedimentation rate (ESR)
- Biochemistry including serum uric acid level
- Prothrombin and partial thromboplastin time
- Cultures of blood, urine or other possible primary sites of infection
- Serological tests for ANA & RF/ACPA
- Radiograph of the joint and contralateral joint
- Synovial fluid analysis or synovial biopsy
- Other imaging studies (Ultrasound, CT or MRI)

Septic arthritis

- A rheumatological emergency
- Requires quick diagnosis and aggressive treatment
- Classically abrupt onset, usually monoarticular involving large weight bearing joints
- Bacterial infections, esp. those due to gm +ve organisms, can destroy the joint cartilage within a few days
- Prompt and proper treatment will usually leave the joint without permanent structural damage.

Septic arthritis

- Haematogenous
- From osteomyelitis
- Adjacent soft tissue infection
- Diagnostic / therapeutic procedures
- Penetrating damage by puncture or cutting

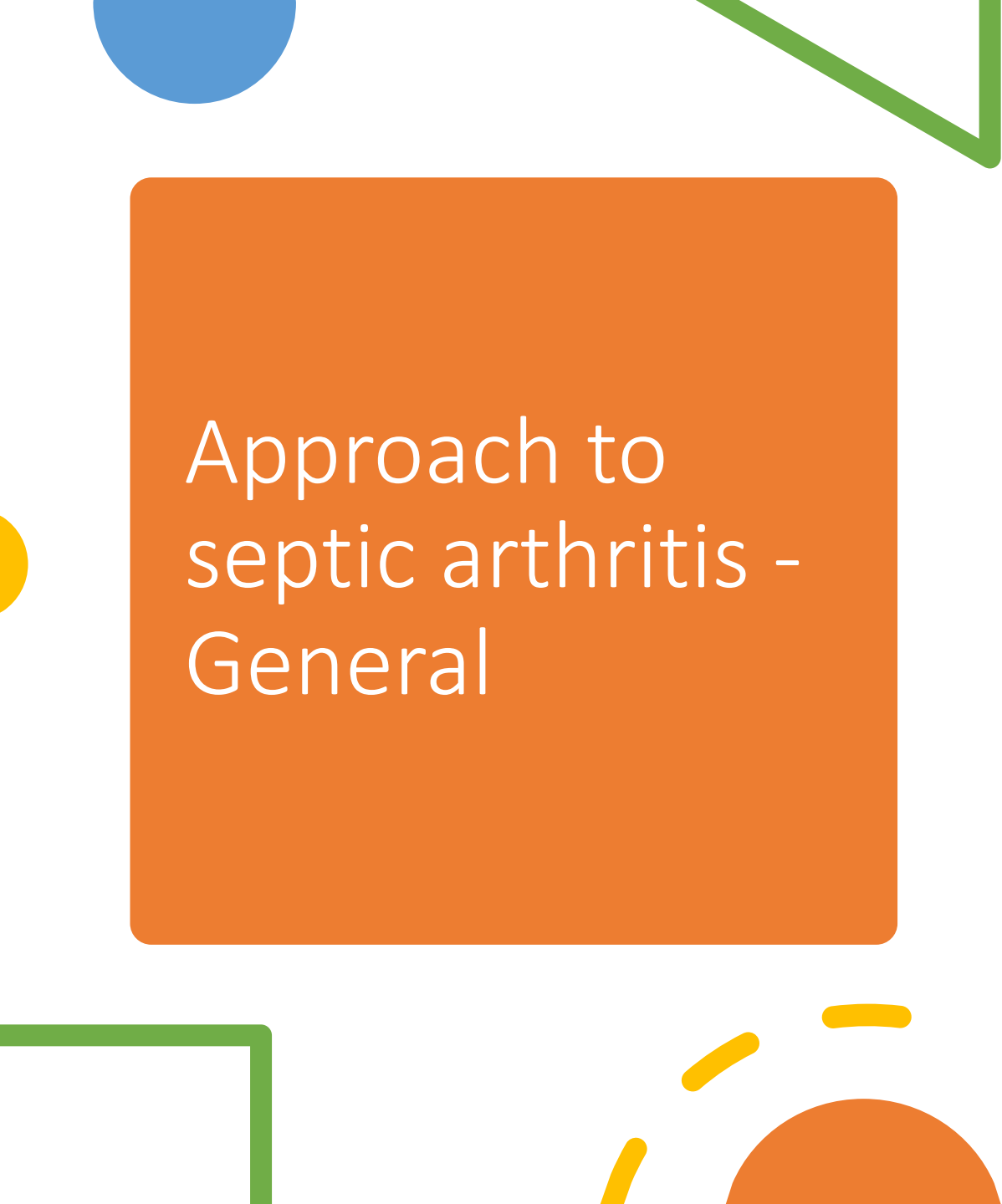


Causes and risk factors of septic arthritis

Causes	Risk factors
Bacterial	• Extreme of age (<5yr or >65yr) • Chronic arthritic syndromes - Rheumatoid arthritis - Crystal induced arthritis - Severe osteoarthritis • Charcot joint or any neuropathic joint • Haemarthroses
Mycobacterial	
Fungal	
Viral	
	• Prosthetic joints • Intra-articular injection or arthrocentesis • Parenteral drug user • Sexual transmitted disease predisposes to localized gonococcal infection, which may disseminate to cause joint disease • Chronic skin infections • Chronic systemic illness - eg. DM, SLE, malignancy, immunocompromised state

Acute bacterial arthritis – special considerations

Special situation	Pathogen
• Infant <1 month	Streptococcal agalactiae Gm-ve bacilli e.g. E. Coli Staphylococcal aureus
• Children <2 years	Haemophilus influenzae type b
• Children >2 years and adults	Staphylococcal aureus
• Elderly and chronically ill	Gm-ve bacilli
• Sexually active young adults	Neisseria gonorrhoeae, particularly of disseminated gonococcal infection. Usually polyarticular
• IV drug abusers	G-ve bacilli, pseudomonas aeruginosa, staphylococcal aureus
• Late prosthetic joint infection	Staphylococcal aureus
• Skin / soft tissue infection	Streptococcal infection
• Foul smelling / gas forming	Anaerobes e.g. clostridium difficile



Approach to septic arthritis - General

- **High level of suspicion**
 - Acute history of a hot, swollen and tender joint(s) with restriction of movement
- **Prompt and appropriate treatment**
 - If suspicion is high, treat even in the absence of fever but only after joint aspiration
 - Negative Gm smear or culture does not exclude sepsis
- **Involve orthopaedic surgeons**



Approach to septic arthritis – investigations

- **Synovial fluid aspiration is key**
 - fluid to be sent fresh for relevant stains and culture
 - prior to commencement of antibiotic therapy
 - warfarin does not contradict needle aspiration
 - suspected hip sepsis requires US guidance
- **Blood cultures should always be taken**
- **CBP, ESR and CRP**
- **Electrolytes and renal and liver function to**
 - to detect end-organ damage
 - guide antibiotic choice
- **Plain x-ray**
- **MRI** is the most appropriate imaging where required
 - Also sensitive in detecting osteomyelitis



Septic arthritis - surgical management

- Confirm diagnosis by examination of joint aspirate
- Urgent joint irrigation and debridement
- Acute phase : arthroscopy
- Delayed or chronic cases : open arthrotomy
- Repeated surgical debridement until infection under control
- Be prepared for co-existing osteomyelitis

Crystal arthritis

Crystalline particles in joint disease

Intrinsic	Extrinsic
• Monosodium urate monohydrate • Calcium pyrophosphate dihydrate • Calcium phosphates - Basic: <i>hydroxyapatite, octacalcium phosphate, tricalcium phosphate</i> - Acidic: <i>brushite, monetite, calcium oxalate</i> • Lipids - Cholesterol - Lipid liquid crystals - Charcot-Leyden (phospholipase) crystals • Cystine • Xanthine, hypoxanthine • Protein precipitates (eg. cryoglobulins)	• Synthetic corticosteroids • Plants thorns (semicrystalloid cellulose), especially blackthorn, rose, dried palm fronds • Sea-urchin spines (crystalline calcium carbonate)

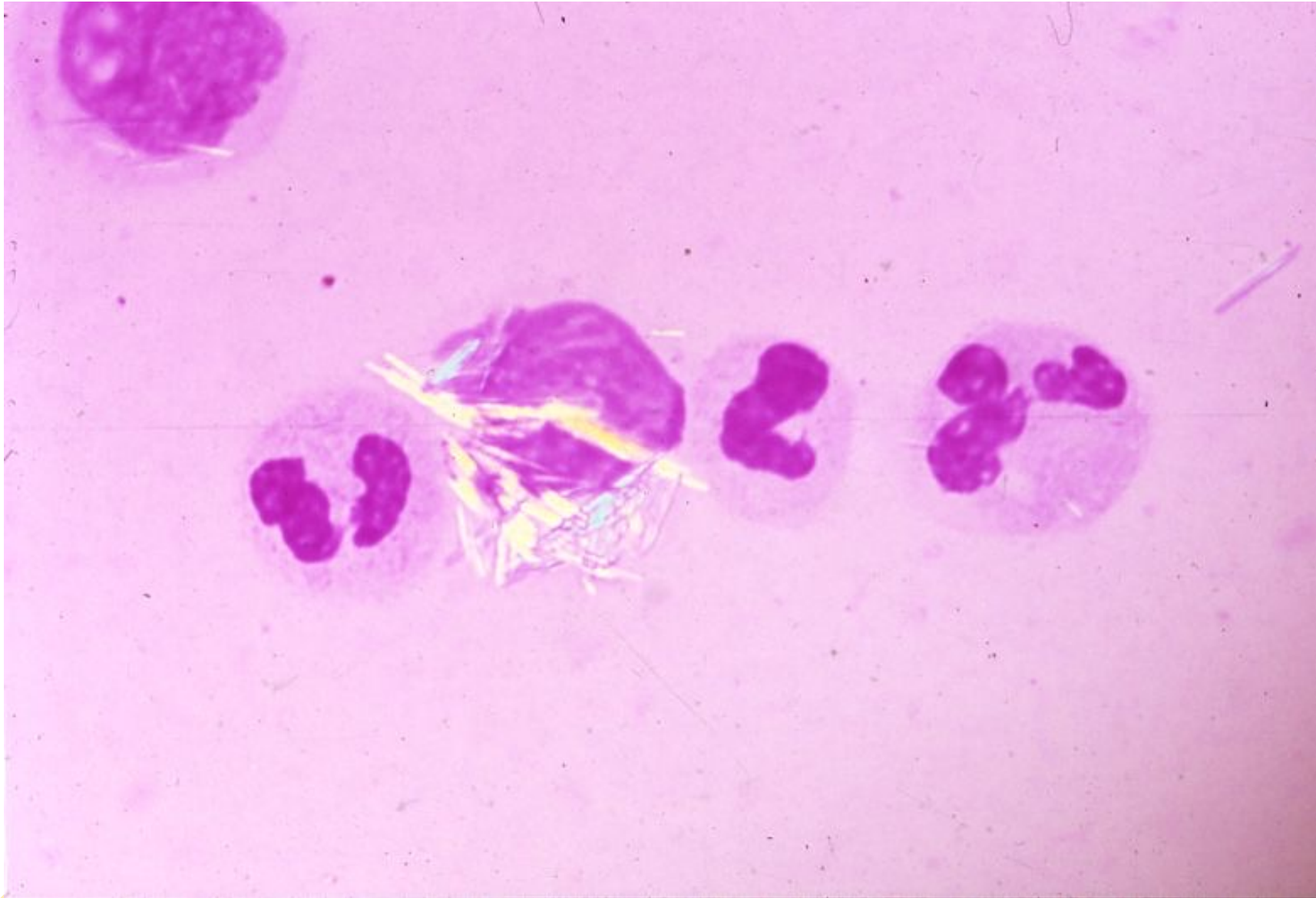
Clinical features of gout

Asymptomatic hyperuricaemia

Acute gouty arthritis

- M:F = 8:1
- Commonest age: 30 to 60 years
- Initial presentation – monoarticular
- First MTP joint affected first in 20%
- Systemic upset is common during acute stage
- Pruritus and desquamation of overlying skin
- Diagnosis by demonstration of UA crystals in joint
- Serum UA level useful but not diagnostic





Clinical features of gout

Intercritical gout

- Entirely asymptomatic in between attacks
- Attacks getting more frequent
- May become polyarticular

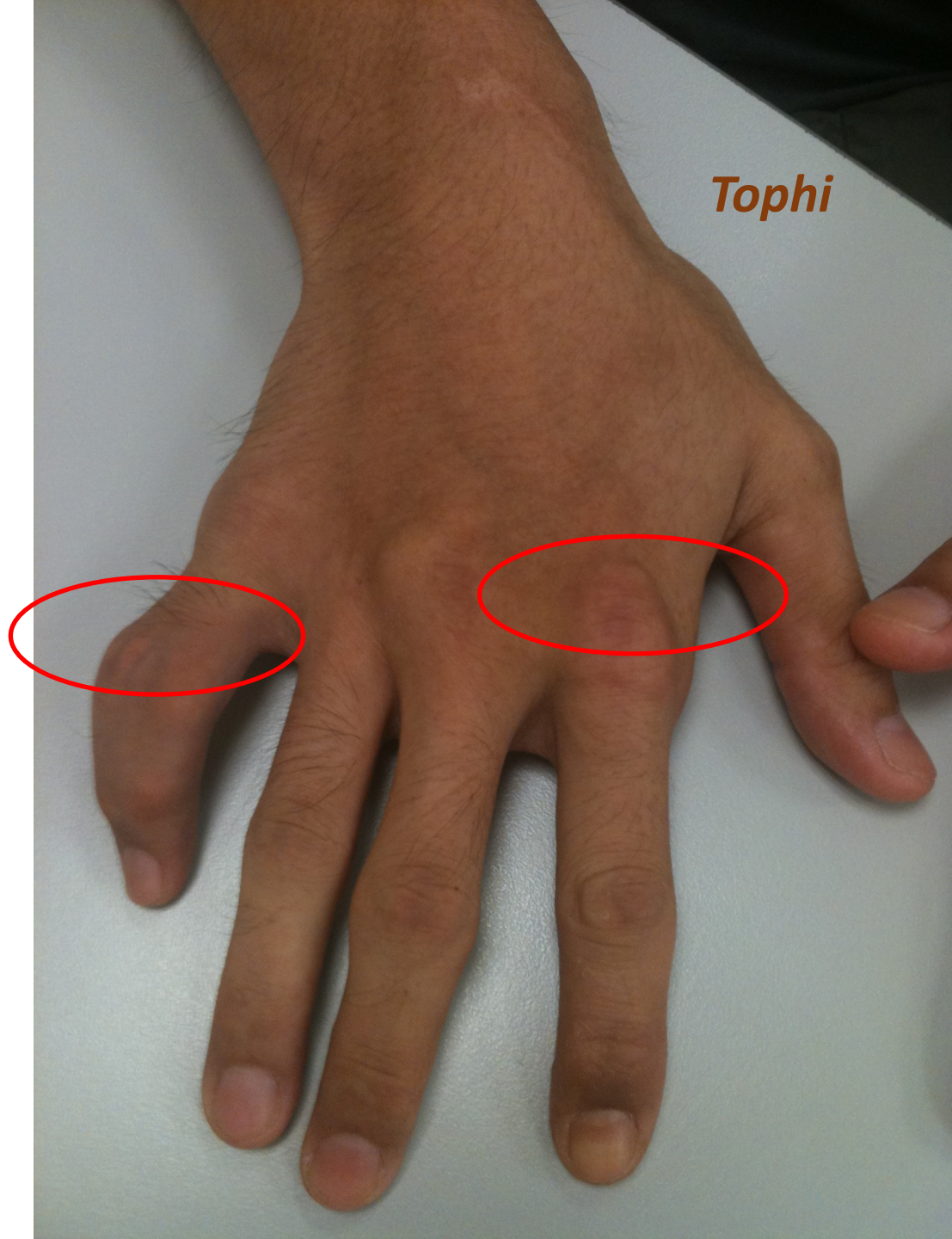
Chronic tophaceous gout

- Tophi in periarticular tissues, helix of ears, tendon sheaths
- Rarely in larynx, tongue, heart
- Occurs in 60% of patients after 10 years

Gout : Clinical associations

- Obesity
- Heavy alcohol intake
- Impaired glucose tolerance
- Ischaemic heart disease
- Hypertension
- Type IV hyperlipoproteinaemia





Tophi

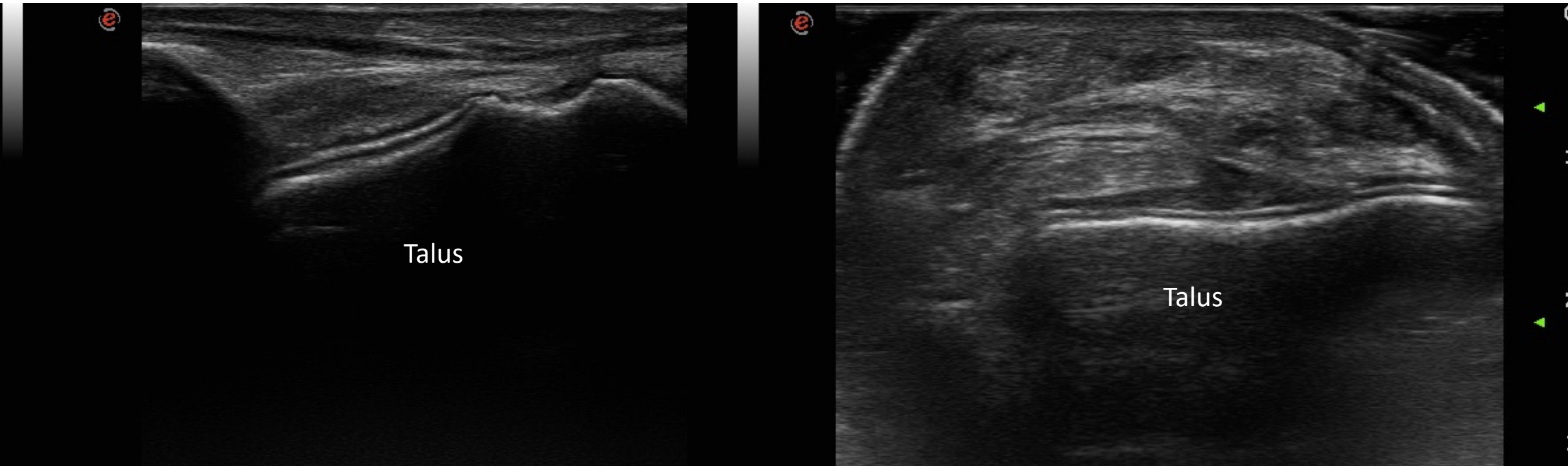


Tophi

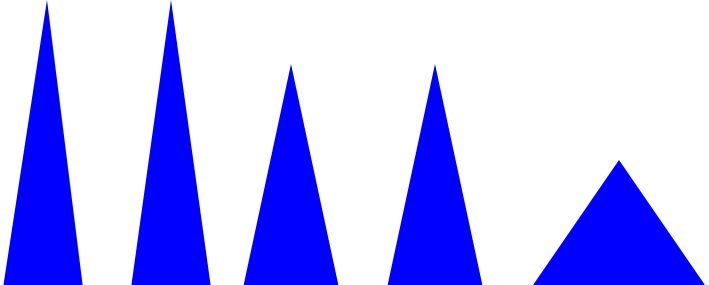
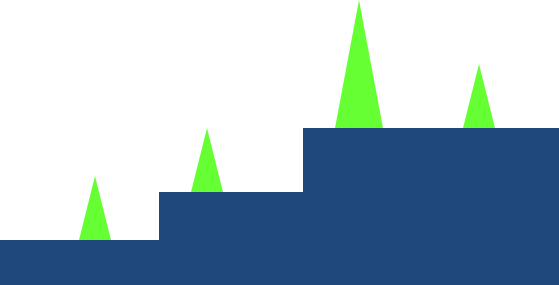


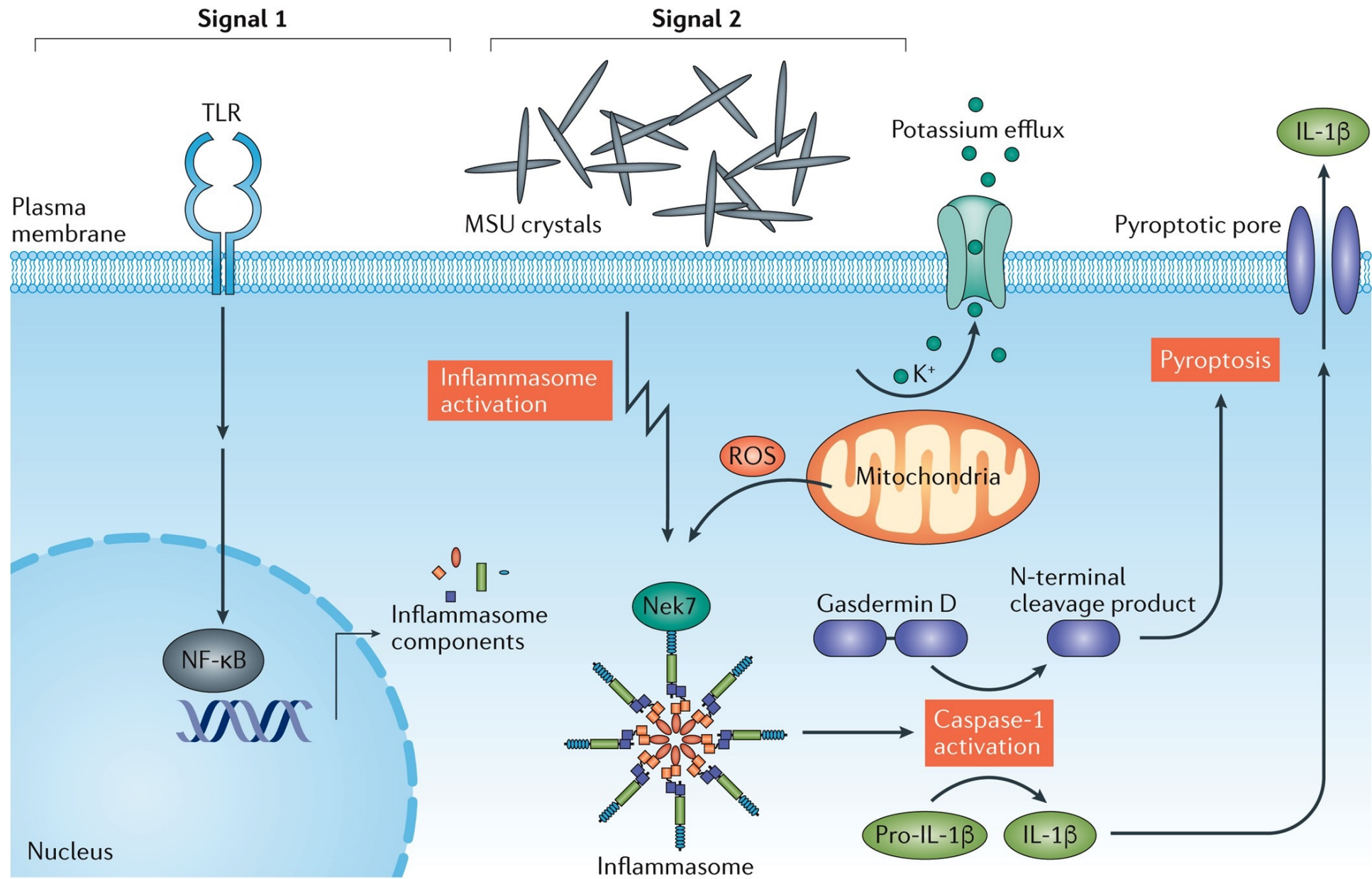


Double contour sign in ultrasound



The natural history of gout

<u>Stage I</u>	<u>Stage II</u>	<u>Stage III</u>
Asymptomatic Hyperuricemia	Acute gouty arthritis and intercritical gout 	Chronic gouty arthritis 
No arthritis	Duration 1-2 wks 1 wk- 2 mos Interval 6 mo-2 yrs 2 wks-4 mos Number of joints involved 1-2 4-5	Continuous arthritis with superimposed acute attack Tophi



Macrophage

Permission from
Nature Reviews | Rheumatology

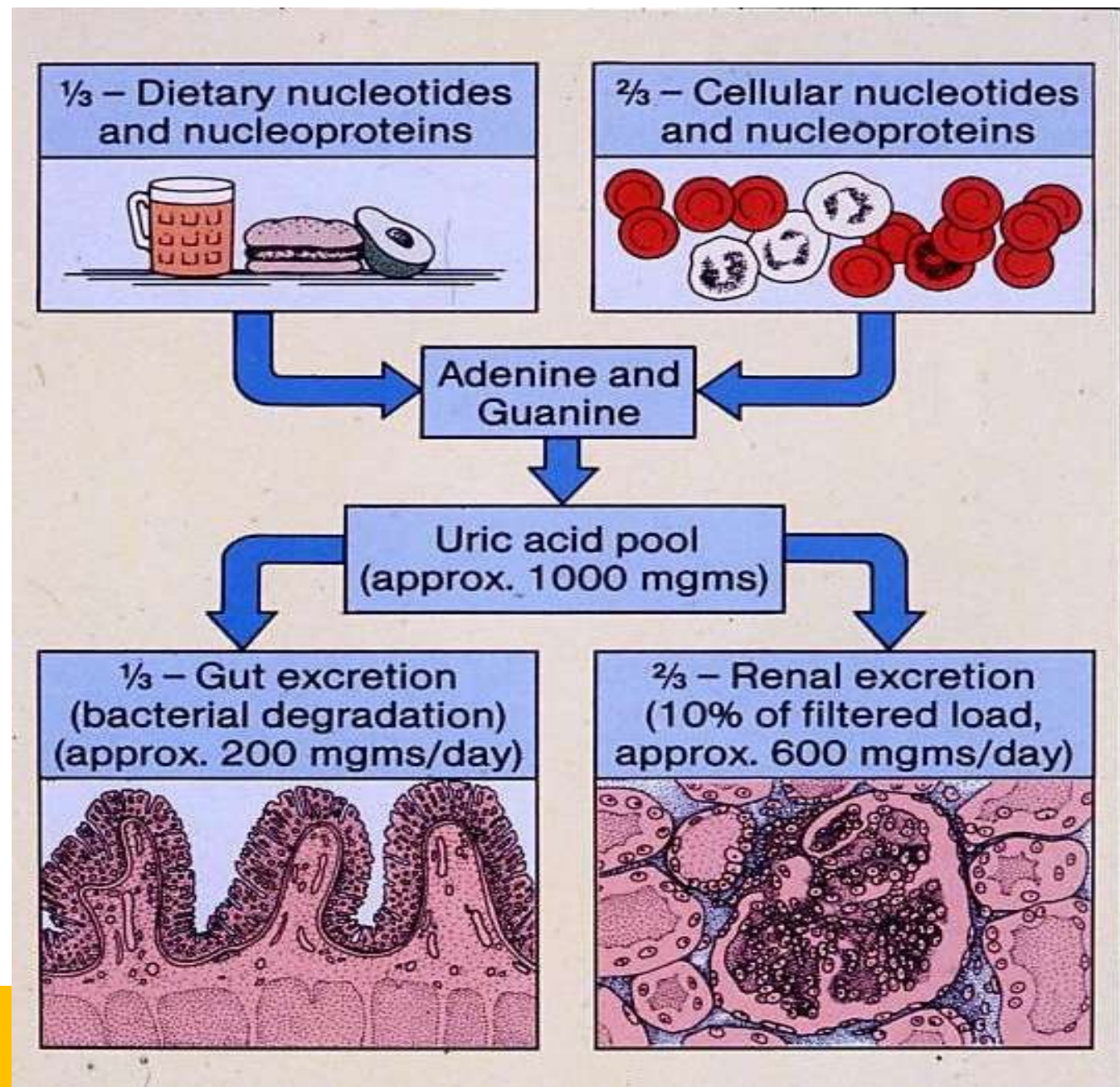
Causes of hyperuricemia

1. Dietary excess

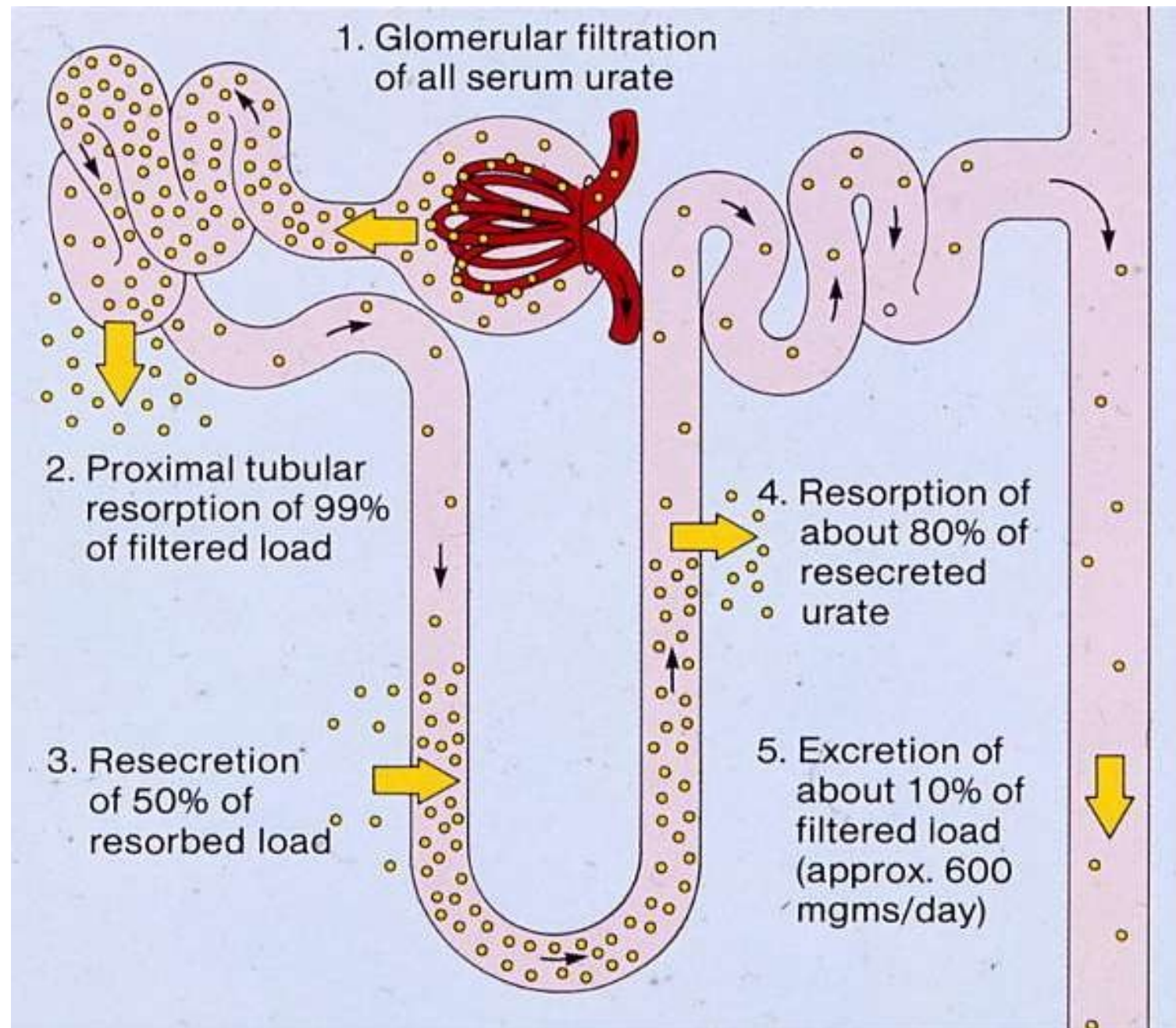
2. Overproduction of urate

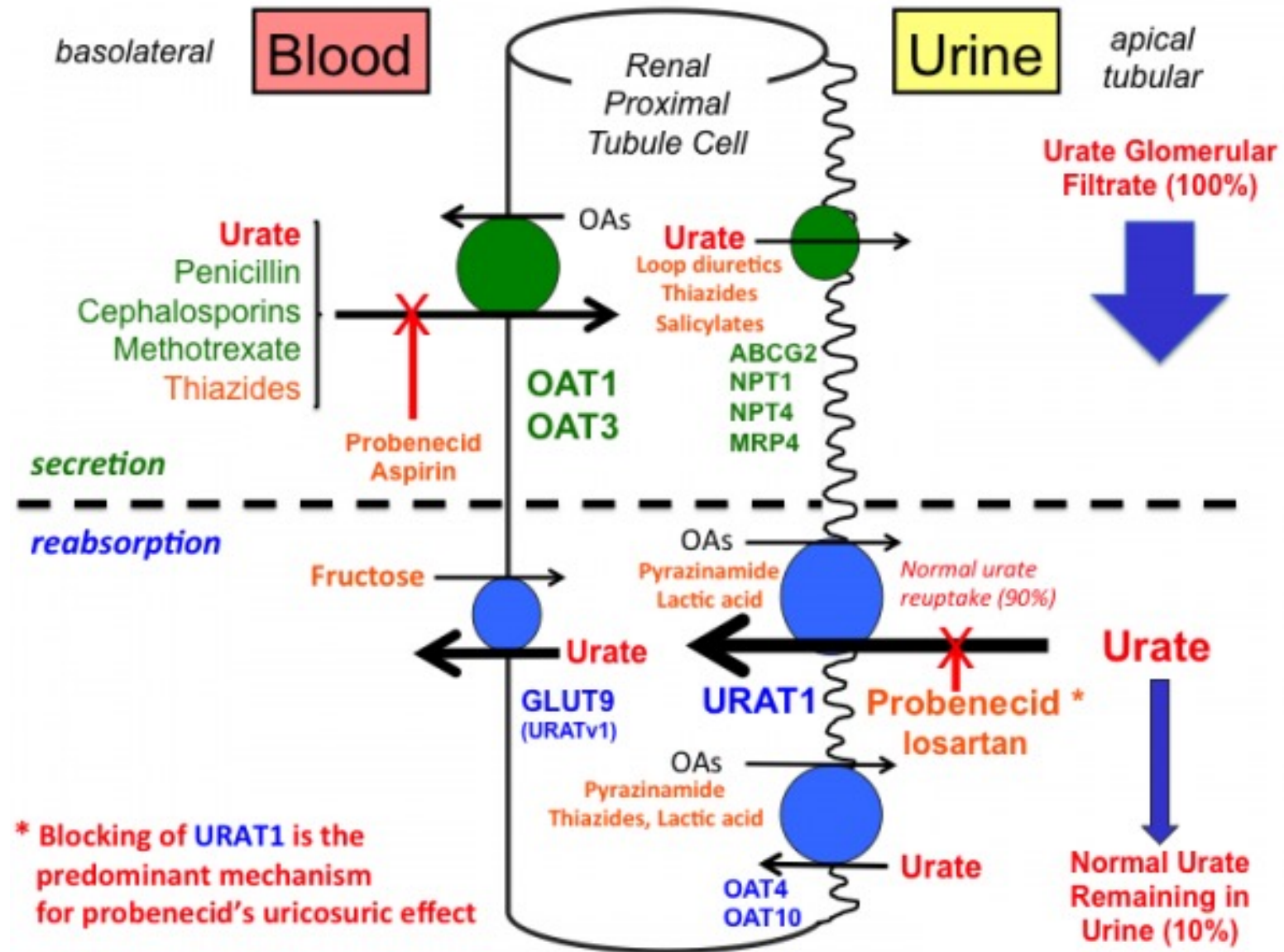
3. Underexcretion of urate

Metabolism pathway of uric acid



The Role of the Intestine in the Development of Hyperuricemia. Front. Immunol; 24 Feb 2022





Overproduction of uric acid

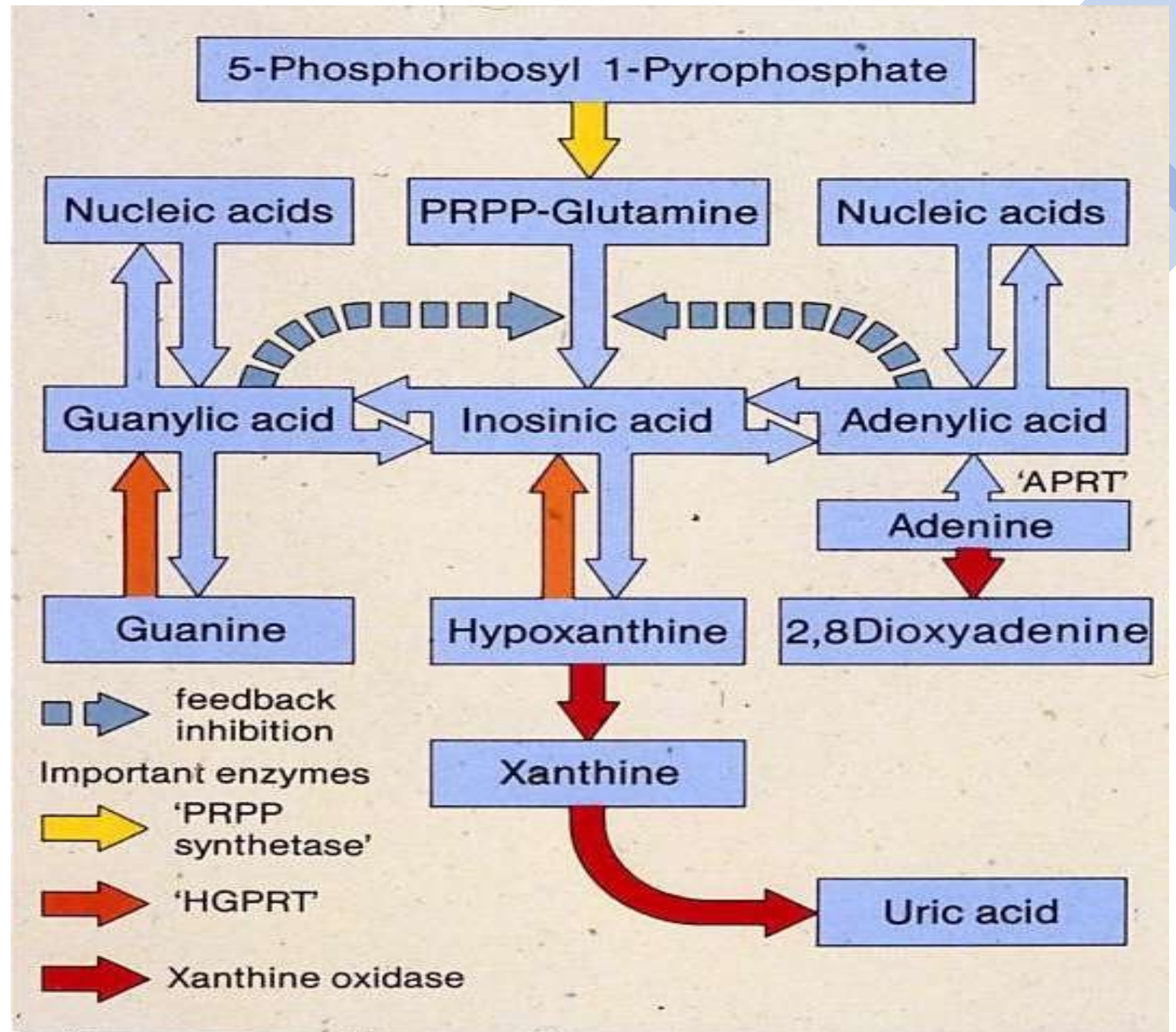
Primary hyperuricaemia

- Idiopathic
- Complete or partial deficiency of HGPRT (hypoxanthine-guanine phosphoribosyl transferase)
- Superactivity of PRPP (phosphoribosyl pyrophosphate synthetase)

Secondary hyperuricaemia

- Excessive purine consumption
- Myeloproliferative or lymphoproliferative disorders
- Haemolytic diseases
- Tumour lysis syndrome
- Psoriasis
- Glycogen storage diseases : types 1, 3, 5 and 7

Purine Salvage pathway



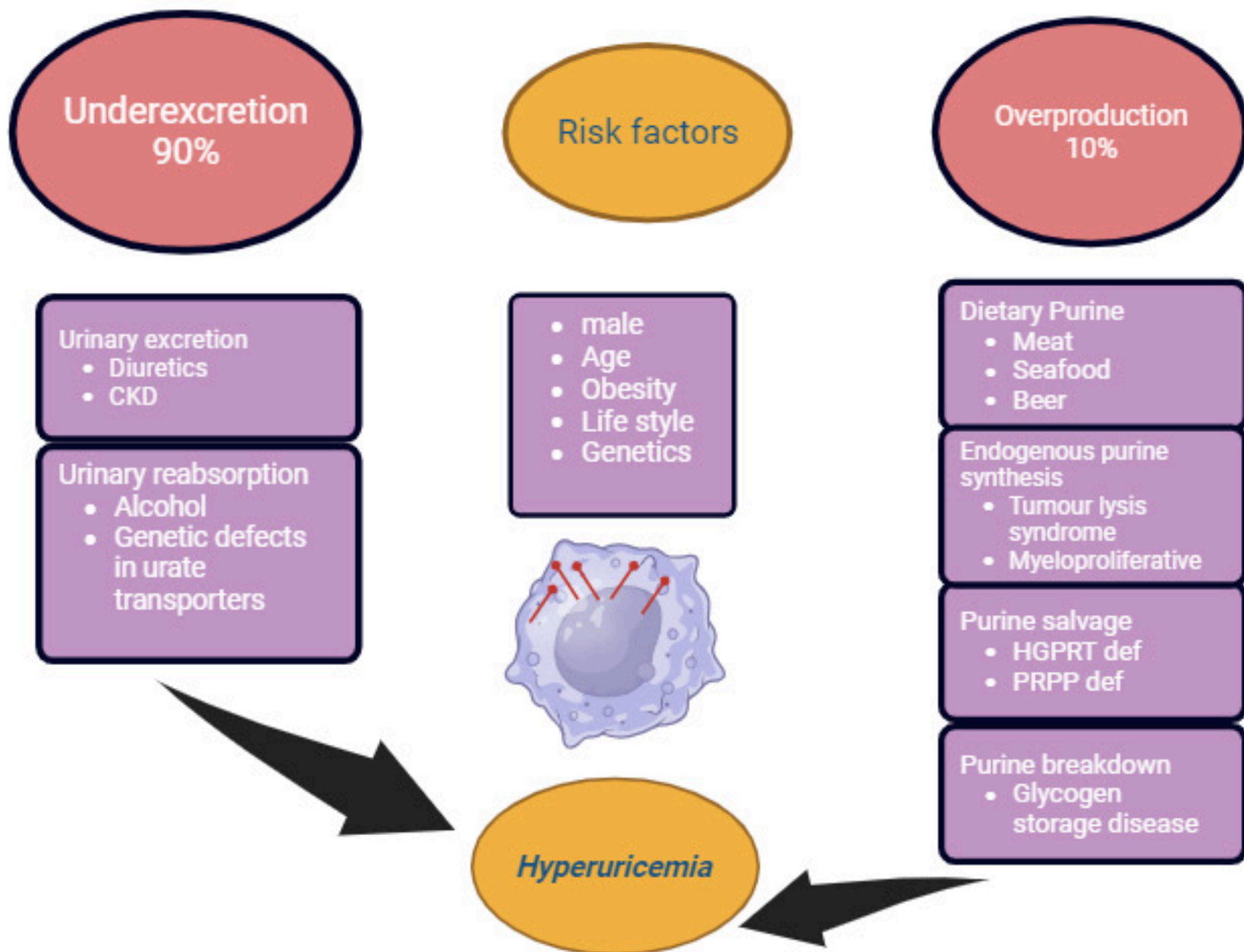
Underexcretion of uric acid

Primary hyperuricaemia

- Idiopathic

Secondary hyperuricaemia

- Chronic kidney disease (CKD)
- Metabolic acidosis (ketoacidosis or lactic acidosis)
- Dehydration
- Diuretics
- Hypertension
- Hyperparathyroidism
- Drugs including cyclosporine, pyrazinamide, ethambutol and low-dose salicylates
- Lead nephropathy



1

Provide **rapid** and
safe pain **relief**

2

Prevent further
attacks

3

Prevent formation
of **tophi and**
destructive
arthritis

4

Address
associated
medical
conditions

Management of gout - objectives



Drugs used in treatment of gout

- Acute attacks
- NSAID : Oral indomethacin is widely used. Aspirin and other salicylates should be avoided
- COX-2 selective inhibitors : a recent study shows etoricoxib to be as effective as indomethacin
- Colchicine : of diagnostic as well as therapeutic value but narrow margin between therapeutic and doses



Drugs used in treatment of gout

- Acute attacks (second line)
- Steroid : Intra-articular, oral or parental
- Anti-IL 1 : Anakinra, Canakinumab (injection)

Drugs used in treatment of gout

Long term management

Allopurinol : a xanthine oxidase inhibitor

Check HLA B5801

usual dose is 300mg daily but higher doses may be used
smaller dose in renal insufficiency

Febuxostat : a xanthine-oxidase inhibitor

Uricosuric agents : probenecid or sulphapyrazone

contraindicated in patients with :

primary overproduction of uric acid

gross hyperuricosuria

evidence of urolithiasis

renal insufficiency

Important Drug Interactions:

Allopurinol or febuxostat should be avoided in patients taking Azathioprine

Newer urate lower agents

- Pegloticase: recombinant uricase
- Lesinurad : URAT1 inhibitor



Long term treatment of gout



Start urate lowering agents ASAP. No need to wait until acute attack has completely subsided

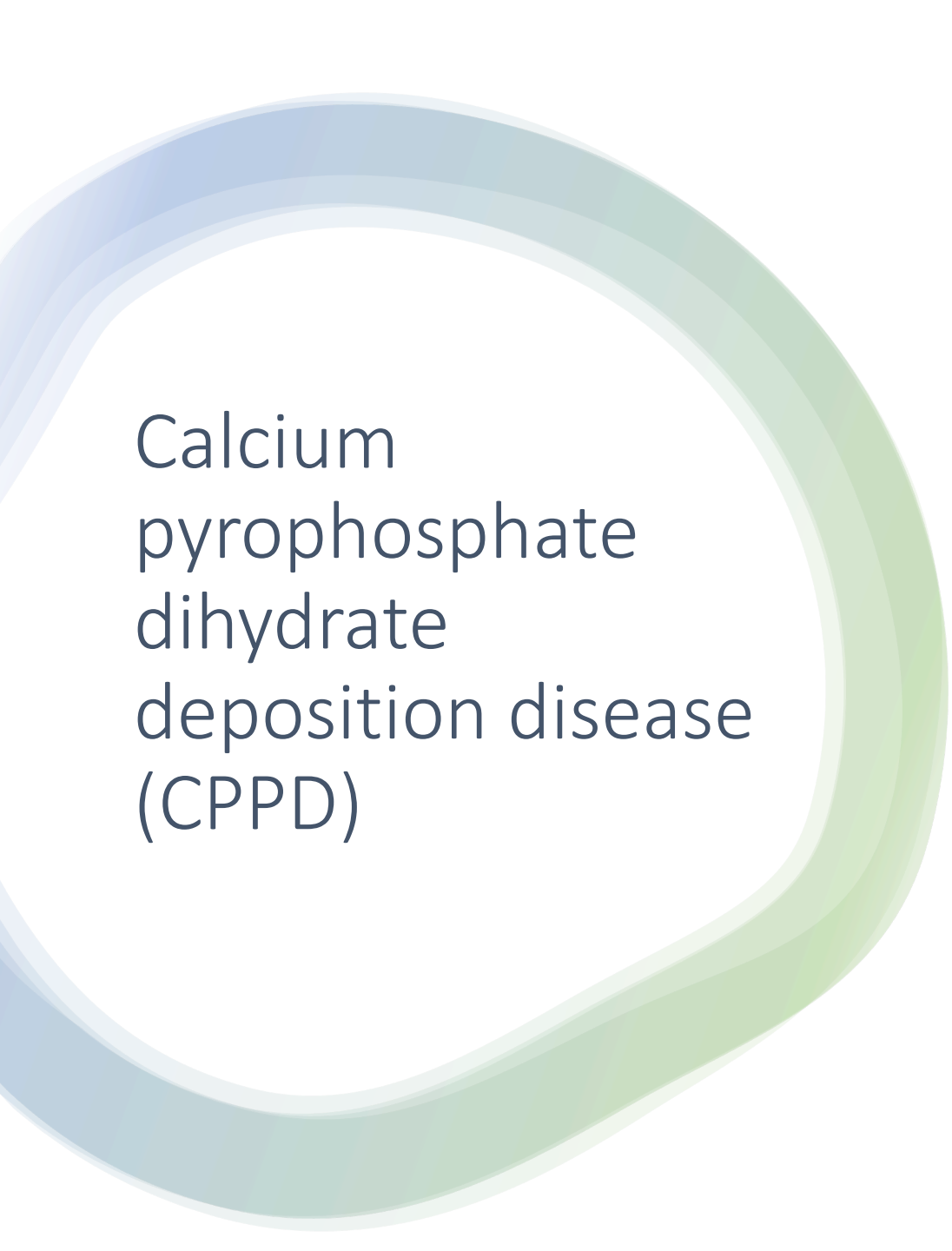


Co-administration with prophylactic colchicine during the first 3 to 6 months to prevent 'breakthrough' attacks



Serious toxic reactions may rarely occur including SCAR in the form of toxic epidermal necrolysis (allopurinol), nephrotic syndrome (probenecid), and hepatitis and bone marrow suppression with both types of hypouricaemic drugs

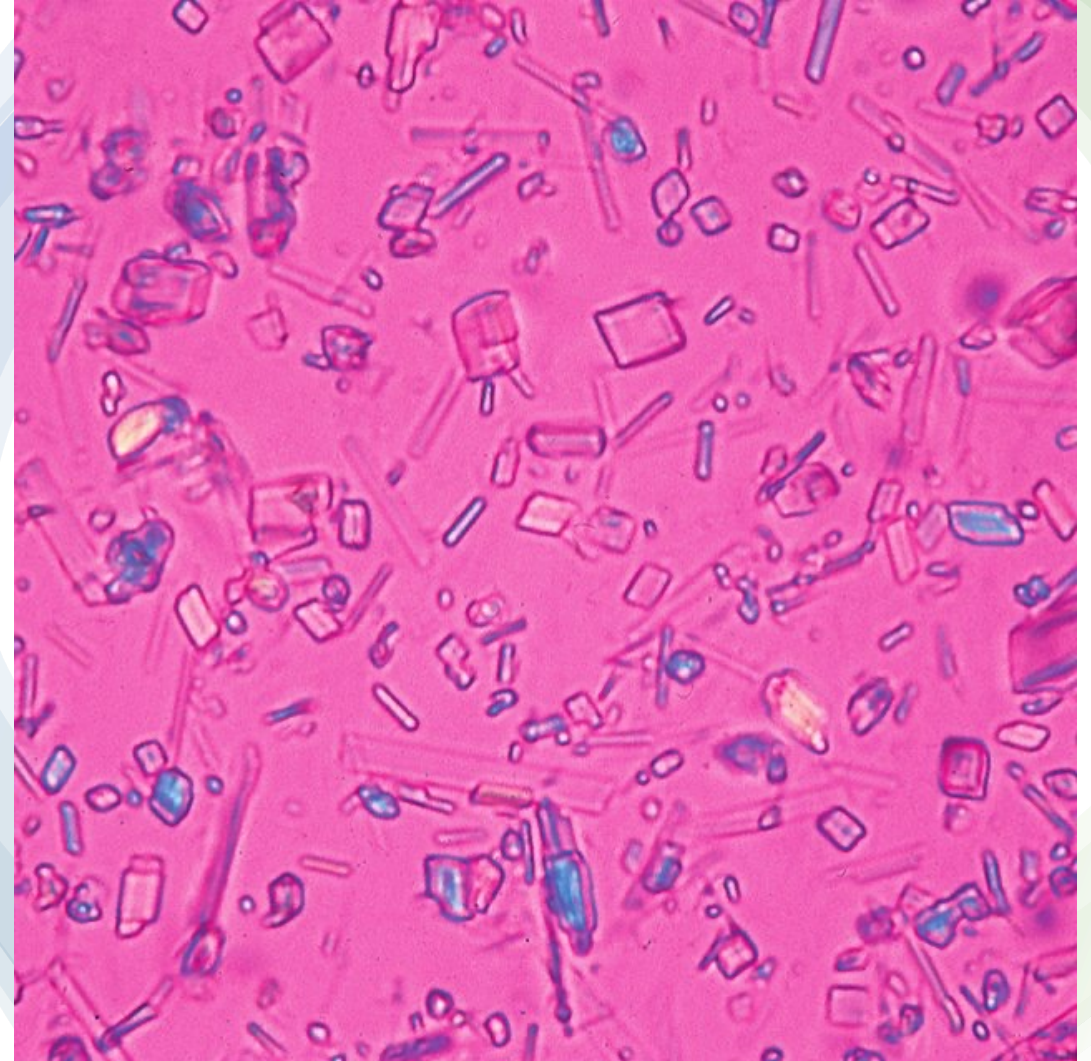
Calcium
pyrophosphate
dihydrate
deposition
disease (CPPD)



Calcium pyrophosphate dihydrate deposition disease (CPPD)

- Synonyms : Chondrocalcinosis, pseudogout
- Present in 6% of individuals >72 yrs old.
 - M:F = 1.4:1.
 - Increases with age
- Incidence of symptomatic disease is about half that of gout
- Most patients have no symptoms at all
- May mimic gout, RA, osteoarthritis, septic arthritis and rarely ankylosing spondylitis
- Knees most commonly affected

CPPD crystal



B

Causes of Pseudogout

Hereditary

Idiopathic – Aging (>65 y.o)

Probable metabolic disease associations :

- hyperparathyroidism
- Haemachromatosis
- Hypothyroidism
- Gout
- Hypomagnesemia
- Hypophosphatasia

Treatment of CPPD disease

Asymptomatic cases do not require treatment

Acute attacks – treat as per gout

Not possible to remove CPPD crystals.

Treatment of underlying associated diseases may not improve long term prognosis

Haemarthrosis

Haemarthrosis

- Common causes
 - Trauma
 - Haemophilia (Congenital VIII and IX def)
 - Acquired haemophilia (autoantibodies)
 - Drugs (warfarin or heparin)
- Symptoms
 - Pain, swelling, stiffness
- Diagnosis: Arthrocentesis
- Treatment: treat the “treatable” underlying causes

Synovial fluid analysis

Gross examination	Normal	Haemarthrosis	Septic
Viscosity	High	Low	Variable
Colour	Straw to yellow	Bloody	Variable to pusy
Clarity	Transparent	Blood like	Opaque
WBC (per mm ³)	<200	Similar to blood level	Often >100,000
PMN leukocytes (%)	<25	Similar to blood level	>75

Diseases causing monoarticular symptoms

Septic

- **Bacterial**
- **Mycobacterial**
- **Lyme disease**

Crystal deposition diseases

- Gout
- Calcium pyrophosphate dihydrate deposition (CPPD) disease (pseudogout)
- Hydroxyapatite deposition disease
- Calcium oxalate deposition disease

Trauma

- Fracture
- ACL/PCL tear
- Haemarthrosis

Others

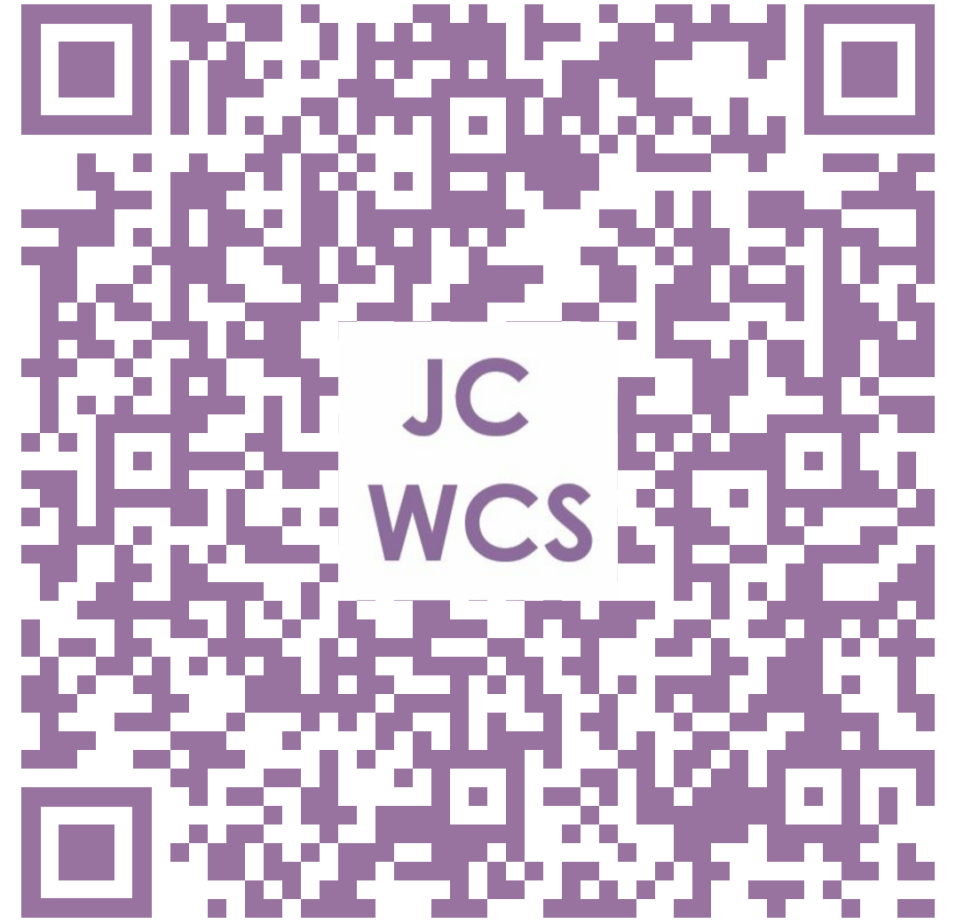
- Osteoarthritis
- Coagulopathy/haemarthrosis
- Avascular necrosis of bone

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.

(HKU Portal login required)